FEMM Magnetics File Formats

Input (`.fem`) and Output (`.ans`) Specifications

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Contents

1 Overview

The FEMM magnetics solver (`fkn`) reads a problem definition from a `.fem` file and writes the solution to an `.ans` file. The `.ans` file begins with a verbatim copy of the input `.fem` file, followed by a `[Solution]` section containing the finite element mesh and computed field values.

All floating-point values are written with 17 significant digits (`%.17g` format) to preserve full double precision across read/write cycles. Fields within a line are separated by tabs or spaces. Section headers use square brackets (`[Tag]`); property block delimiters use angle brackets (`<BeginBlock>` / `<EndBlock>`). String comparisons during parsing are case-insensitive.

2 The `.fem` Input File

2.1 Global Problem Parameters

Each parameter occupies one line in the format `[Tag] = value`. All are optional; defaults are applied if missing.

Tag	Type	Default	Description
`[Format]`	double	4.0	File format version
`[Frequency]`	double	0	AC frequency (Hz); 0 = DC
`[Precision]`	double	1e-8	Solver convergence tolerance
`[MinAngle]`	double	30	Minimum mesh element angle (degrees)
`[DoSmartMesh]`	int	1	Adaptive meshing (0=off, 1=on)
`[Depth]`	double	1.0	Extrusion depth (planar problems)
`[LengthUnits]`	string	millimeters	inches, millimeters, centimeters, meters, mils, microns
`[ProblemType]`	string	planar	planar or axisymmetric
`[Coordinates]`	string	cartesian	cartesian or polar
`[ACSolver]`	int	0	0=successive approximation, 1=Newton
`[PrevType]`	int	0	0=none, 1=incremental, 2=frozen, 3=restart
`[PrevSoln]`	string	(empty)	Path to previous solution (quoted)
`[Comment]`	string	(empty)	Problem description (quoted; `\n` = newline)

2.1.1 Axisymmetric External Region

Written only when `ProblemType` is axisymmetric and both `extRo` and `extRi` are nonzero:

Tag	Type	Description
`[extZo]`	double	Outer Z coordinate
`[extRo]`	double	Outer radius
`[extRi]`	double	Inner radius

2.2 Point (Nodal) Properties

```
[PointProps] = <count>
<BeginPoint>
  <PointName> = "name"
  <I_re> = 0      real part of point current (A)
  <I_im> = 0      imaginary part (AC only)
  <A_re> = 0      real part of prescribed A
  <A_im> = 0      imaginary part (AC only)
<EndPoint>
... repeated count times
```

2.3 Boundary Properties

```
[BdryProps] = <count>
<BeginBdry>
  <BdryName> = "name"
  <BdryType> = 0
  <A_0> = 0        <A_1> = 0        <A_2> = 0
  <Phi> = 0
  <c0> = 0         <c0i> = 0
  <c1> = 0         <c1i> = 0
  <Mu_ssd> = 0     <Sigma_ssd> = 0
  <innerangle> = 0 <outerangle> = 0
<EndBdry>
```

Boundary types:

- 0 = prescribed A
- 1 = small skin depth
- 2 = mixed
- 3 = strategic dual image (SDI)
- 4 = periodic
- 5 = antiperiodic
- 6 = periodic air gap element (AGE)
- 7 = antiperiodic air gap element (AGE)

2.4 Material (Block) Properties

```
[BlockProps] = <count>
<BeginBlock>
  <BlockName> = "name"
  <Mu_x> = 1       relative permeability, x-direction
  <Mu_y> = 1       relative permeability, y-direction
  <H_c> = 0        coercivity (A/m)
  <H_cAngle> = 0   magnetization direction (degrees)
  <J_re> = 0       source current density, real (MA/m^2)
  <J_im> = 0       source current density, imag (MA/m^2)
  <Sigma> = 0      conductivity (S/m)
  <d_lam> = 0      lamination thickness (mm)
  <Phi_h> = 0      hysteresis lag angle (degrees)
  <Phi_hx> = 0     hysteresis lag angle, x (degrees)
```

```
  <Phi_hy> = 0     hysteresis lag angle, y (degrees)
  <LamType> = 0    lamination type
  <LamFill> = 1    lamination fill factor (0--1)
  <NStrands> = 0   number of wire strands
  <WireD> = 0      wire diameter (mm)
  <BHPoints> = N
    B_0  H_0       B-H curve data (Tesla, A/m)
    B_1  H_1       ... repeated N times
    ...
<EndBlock>
```

Lamination types:

- 0 = not laminated
- 1 = laminated in x (parallel to x)
- 2 = laminated in y (parallel to y)
- 3 = magnet wire
- 4 = plain stranded
- 5 = litz wire
- 6 = rectangular wire
- 7 = 10% CCA
- 8 = 15% CCA

2.5 Circuit Properties

```
[CircuitProps] = <count>
<BeginCircuit>
  <CircuitName> = "name"
  <TotalAmps_re> = 0   total current, real (A)
  <TotalAmps_im> = 0   total current, imag (A)
  <CircuitType> = 0    0=series, 1=parallel
<EndCircuit>
```

2.6 Geometry

2.6.1 Nodes

```
[NumPoints] = <count>
x    y    propIdx   inGroup   [lfs]
```

- `x`, `y`: coordinates (double)
- `propIdx`: index into point property list (1-based; 0 = none)
- `inGroup`: group number (int)
- `lfs`: local feature size constraint (double, optional; -1 = none)

2.6.2 Line Segments

```
[NumSegments] = <count>
n0    n1    maxSideLength   propIdx   hidden   inGroup
```

- `n0`, `n1`: endpoint node indices (0-based)

- `maxSideLength`: mesh size constraint (double; -1 = auto)
- `propIdx`: boundary property index (1-based; 0 = none)
- `hidden`: 0 = visible, 1 = suppressed from mesh
- `inGroup`: group number

2.6.3 Arc Segments

```
[NumArcSegments] = <count>
n0  n1  arcAngle maxSideLength propIdx hidden inGroup mySideLength
```

- `n0`, `n1`: endpoint node indices (0-based)
- `arcAngle`: arc span (degrees)
- `maxSideLength`: mesh constraint (double)
- `propIdx`: boundary property index (1-based; 0 = none)
- `hidden`: 0 = visible, 1 = suppressed
- `inGroup`: group number
- `mySideLength`: computed side length (0 = auto)

2.6.4 Holes

```
[NumHoles] = <count>
x    y    inGroup
```

Holes are seed points for regions excluded from the mesh. They correspond to block labels with material `"<No Mesh>"`.

2.6.5 Block Labels

```
[NumBlockLabels] = <count>
x  y  blockIdx meshSize circIdx magDir inGroup turns extFlags ["magDirFctn"]
```

- `x`, `y`: label position (double)
- `blockIdx`: material index (1-based; 0 = unassigned)
- `meshSize`: element size (double; ≤ 0 = auto). Converted to area: $A = \pi r^2/4$
- `circIdx`: circuit index (1-based; 0 = none)
- `magDir`: magnetization direction (degrees)
- `inGroup`: group number
- `turns`: number of turns
- `extFlags`: bit 0 = external region, bit 1 = default region
- `magDirFctn`: magnetization direction function (optional, quoted string)

3 The `.ans` Output File

The `.ans` file begins with a verbatim copy of the input `.fem` file (all sections described in Section 2), followed by the solver's solution data.

3.1 Solution Header

Immediately after the echoed `.fem` content:

```
[Solution]
```

3.2 Mesh Nodes

```
<NumNodes>
```

followed by `NumNodes` lines. The format depends on the problem type:

DC (Frequency = 0), no previous solution:

```
x    y    A_re
```

DC with incremental/frozen permeability:

```
x    y    A_re    bc    A_prev
```

AC (Frequency > 0), no previous solution:

```
x    y    A_re    A_im
```

AC with incremental/frozen permeability:

```
x    y    A_re    A_im    bc    A_prev
```

- `x`, `y`: node coordinates in model units
- `A_re`, `A_im`: magnetic vector potential (Wb/m)
- `bc`: boundary condition index (-1 = none)
- `A_prev`: vector potential from previous solution

Node indices are 0-based throughout the solution section.

3.3 Mesh Elements

```
<NumElements>
```

followed by `NumElements` lines:

```
p0    p1    p2    labelIdx    e0    e1    e2    J_prev
```

- `p0`, `p1`, `p2`: vertex node indices (0-based)
- `labelIdx`: block label index (0-based, into the block label list from the `.fem` header)
- `e0`, `e1`, `e2`: boundary property index applied to each of the element's three edges, or -1 where the edge carries no boundary condition
- `J_prev`: current density carried over from the previous solution, in $\mathbf{MA/m^2}$ — the solver's working unit, not the A/m^2 that the postprocessor reports. Consumers must scale by 10^6 to combine it with current densities obtained from the postprocessor.

The static solver always writes all eight fields. The harmonic solver writes the first seven and appends `J_prev` only for an incremental (frozen-permeability) run.

Note on `J_prev`'s position. It is the *eighth* field, after the three edge indices. Original FEMM's postprocessor reads it as the fifth — picking up `e0`, a boundary index that is -1 on every interior element — which corrupts its $1\times$ incremental Lorentz force and torque integrals (block integral types 28–30). femm-qt reads the field the solver actually writes, so those three integrals differ from original FEMM by design; every other quantity is unaffected.

Elements are written in Cuthill-McKee order (sorted by sum of node indices), which clusters spatially nearby elements at nearby indices for efficient postprocessor lookup.

3.4 Circuit Solution Data

```
<NumCircuits>
```

followed by `NumCircuits` lines (one per block label in the problem):

DC:
```
caseType   value_re
```

AC:
```
caseType   value_re   value_im
```

- `caseType`:
  - 0 = voltage gradient (value = ΔV, voltage drop)
  - 1 = current density (value = J)
  - 2 = flux constraint (from solver's flux equation)
- `value_re`, `value_im`: solution values for the circuit associated with this block label

3.5 Periodic Boundary Conditions

```
<NumPBCs>
```

followed by `NumPBCs` lines:

```
nodeIdx0   nodeIdx1   type
```

- `nodeIdx0`, `nodeIdx1`: paired node indices (0-based)
- `type`: 0 = periodic, 1 = antiperiodic

3.6 Air Gap Elements

```
<NumAGEs>
```

For each air gap element:

Line 1: Boundary name (quoted string)

```
"BoundaryName"
```

Line 2: Parameters

```
bdryFmt innerAngle outerAngle ri ro  totalArc agc_re agc_im numArcElems innerShift
    outerShift
```

- `bdryFmt`: 0 = periodic, 1 = antiperiodic
- `innerAngle`, `outerAngle`: boundary angles (degrees)
- `ri`, `ro`: inner and outer radii (model units)
- `totalArc`: total arc length (degrees)
- `agc_re`, `agc_im`: air gap coefficient (complex)
- `numArcElems`: number of arc elements
- `innerShift`, `outerShift`: boundary shifts

Lines 3 to $2 + \mathbf{numArcElems} + 1$: Quadrature points

```
n0  w0  n1  w1  n2  w2  n3  w3
```

- `n0`–`n3`: node indices for interpolation
- `w0`–`w3`: interpolation weights

3.7 Air Gap Ring Node Lists (optional)

The quadrature-point mapping above is sorted at the rotor angle of the solution that wrote it. When air gap elements are present, the solver appends a trailing section carrying each element's raw ring node lists, which are independent of the rotor angle:

```
[AgeRings]
```

For each air gap element, a count line equal to `numArcElems`, followed by that many lines:

```
innerNodeIdx outerNodeIdx
```

listing the mesh nodes (0-based) on the inner and outer gap rings of the meshed sector. A transient analysis uses these to reuse the previous solution's mesh while rebuilding the quadrature-point pairing at the current step's rotor angle. The section is strictly trailing so that readers which parse the air gap elements positionally and stop are unaffected; when it is absent, the solver falls back to remeshing.

4 Index Conventions

- **Property indices** (materials, boundaries, circuits, point properties) are **1-based** in the file. An index of 0 means "no property assigned." The parser converts to 0-based internally.
- **Node indices** (in segments, arcs, and the solution section) are **0-based**.
- **Block label indices** in the solution element list are **0-based**, referencing the block label array from the `.fem` header.

5 Unit Conversion

The `[LengthUnits]` tag determines the coordinate system. All coordinates in the file are in the specified units. Conversion to meters:

Unit	Meters per unit
inches	0.0254
millimeters	0.001
centimeters	0.01
meters	1.0
mils	2.54×10^{-5}
microns	10^{-6}